Jagan Kumar Swain

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate (CGPA 9.1) with hands-on experience across embedded hardware, drone systems, FPGA/Verilog design, machine learning, and full-stack development. Completed a Micro Drone internship at PE Geometry and a deep learning research internship at IISER Berhampur. Award-winning builder: Best Innovation Award (Chakravyuh 1.0) and Rs. 50,000 Defence R&D Grant for GuardX Sentinel.

EDUCATION

B.Tech, Electronics and Communication Engineering 2024 – 2028
NIST University, Berhampur, Odisha *CGPA: 9.1*

BS, Data Science and Applications 2025 – 2028
Indian Institute of Technology (IIT) Madras *CGPA: 8.1*

EXPERIENCE

Research Intern – Deep Learning for Medical Imaging Jun 2026 – Present
IISER Berhampur, Odisha (Dept. of Computer Science) *Supervisor: Dr. Anabik Pal*

- Proposed and received supervisor approval for the project *Hybrid Deep Learning Framework for Multi-Retinal Disease Localization, Segmentation and Classification from Fundus Images*.
- Developing an integrated deep learning pipeline for detection, localization, segmentation, and classification of multiple retinal diseases from fundus images.
- Compiled candidate public fundus datasets and set up a private GitHub repository with a pull-request workflow for version-controlled team collaboration.
- Reported progress and gathered technical feedback through scheduled supervisor review meetings.

Engineering Intern – Micro Drone Project 05 Jun 2026 – 31 Jul 2026
PE Geometry India Pvt Ltd, Bengaluru, Karnataka *Geometry Assurance Engineering*

- Contributed to the development of a micro drone, working across the mechanical and electronics aspects of the product.
- Applied the product design process to integrate mechanical and electronic subsystems for optimal product functionality.
- Recognized in the internship certificate for quick learning, hard work, and initiative in knowledge exploration.

TECHNICAL SKILLS

Programming Languages: C, Python, Java, JavaScript, HTML, CSS, Verilog
AI / Machine Learning: Machine Learning, Deep Learning, Neural Networks, CNN, LSTM, NLP, PyTorch, Computer Vision, Medical Image Analysis
Embedded & Hardware: Arduino, Raspberry Pi, IoT, FPGA, VLSI, Digital Signal Processing, Drone Systems, EDA Tools
Web & Backend: Node.js, Express, MongoDB, Flask, REST APIs, JWT Authentication
Tools & Platforms: Linux, Git, GitHub, Vivado, MATLAB, Google Colab, Tableau, AWS, Google Cloud Platform, Vertex AI

PROJECTS

GuardX Sentinel – Drone-Based Human Activity Detection System
Computer Vision, IoT, Embedded Systems

- Developed an indigenous drone-integrated surveillance system for real-time human activity detection and threat identification in defense scenarios.
- Integrated Arduino Nicla Vision, thermal and regular camera modules, and onboard sensors for edge processing and real-time data acquisition.
- Optimized the system for low-latency operation under constrained computational resources.
- Won **Best Innovation Award** at Chakravyuh 1.0 and a **Rs. 50,000 Defence R&D Grant** at the International Conference on Air Defence and Security.

SegroBin – Smart Waste Management System (Smart India Hackathon 2025)
IoT, Embedded Systems, Application Development

- Designed a smart waste segregation system for Government of Odisha problem statement SIH25046; selected among the Top 5 finalists for the problem statement.
- Built an IoT-enabled smart dustbin with a companion app for real-time waste tracking and a Green Credits incentive mechanism for recyclable segregation.
- Engineered a low-cost, scalable solution suited to municipal collection and recycling workflows.

DeepBeat – ECG Arrhythmia Classification System
Deep Learning, Medical Time-Series Analysis

- Built a hybrid CNN–LSTM model in PyTorch for cardiac arrhythmia detection, trained on the MIT-BIH Arrhythmia dataset

(87,000+ ECG samples).
- Achieved 96% overall accuracy with high recall on critical ventricular and supraventricular classes.
- Designed as a safety-first clinical screening support tool.

Formula 1 Race Winner Prediction System
Machine Learning, *Time-Series Analysis*
- Engineered features (driver form, season momentum, circuit affinity, grid position) from historical Formula 1 data sourced from Kaggle.
- Compared Logistic Regression, Random Forest, Gradient Boosting, and Neural Network models; achieved up to 95.6% cross-validated accuracy with the Neural Network using time-aware validation.

16-Operation ALU Design using Verilog HDL
Verilog, *Digital Design*, *VLSI*
- Designed a 16-operation ALU in Verilog covering arithmetic (+, -, *, /), logical (AND, OR, XOR), shift, and comparison operations, with Carry and Zero status flags.
- Verified functionality and timing through waveform simulation in Vivado 2025.2.

AI Career Agent – IndiaAI Impact GenAI Hackathon (IISc Bengaluru, MeitY)
AI Agents, *Full-Stack Development*
- Built an AI career guidance agent with skill extraction, job-role matching, and personalized learning path generation.
- Developed a modular architecture with a Flask backend, AI reasoning engine, and single-page frontend; used IBM open-source technologies for agentic workflows beyond keyword matching.

Full-Stack Authentication System
Node.js, *Express*, *MongoDB*
- Implemented secure authentication with JWT, bcrypt password hashing, and input validation, plus a responsive frontend with real-time form validation.

Smart Blind Stick | Arduino-Based 3D Scanner
Embedded Systems, *IoT*, *Computer Vision*
- Smart Blind Stick: built an Arduino-based assistive device with sensor-driven obstacle detection and real-time alerts.
- 3D Scanner: designed a stepper-motor rotating platform synchronized with camera image acquisition for 3D reconstruction.

ACHIEVEMENTS
- **Rs. 50,000 Defence R&D Grant**, Army AD College and NIST University, International Conference on Air Defence and Security (GuardX Sentinel).
- **Best Innovation Award**, Chakravyuh 1.0 (GuardX Sentinel).
- **Grand Finalist**, Smart India Hackathon 2025, Hardware Edition (Team CtrlOhm).
- **1st Place**, Web Development Competition, Festronix 2K26, GIFT Autonomous, Bhubaneswar.
- **Winner**, Code Crusade 4.0 (1st and 2nd Year category), Club Excel, NIST University.

CERTIFICATIONS AND TRAINING
- **Build and Deploy Machine Learning Solutions on Vertex AI**, Google Cloud Skill Badge.
- **Data Analytics Job Simulation**, Deloitte Australia (Forage): data analysis dashboards in Tableau and insights in Excel.
- **Internet of Things Internship**, LaunchED Global: sensors, actuators, Arduino Cloud, Blynk, TinkerCAD, OTA updates, Wi-Fi IoT communication.
- **IoT and Mechatronics Internship**, NIST University: Arduino-based embedded systems, sensors, automation, cloud-integrated IoT.
- **OOSC 3.0 – Open-Source Software Conference**, IIT Kanpur (Canonical and Ubuntu): Linux and open-source development.
- **DevOps Webinar**, Malla Reddy Engineering College and Google Developer Groups: DevOps principles and CI/CD pipelines.
- **International Tech Event**, Google and Google Developer Groups: Vertex AI and Gemini applications.
- **Workshops**, NIST University: SensoTech (IoT sensors, RFID attendance and rain-alert mini-projects); Blender and AR (Blender, Unity); AI Cloudwave (AWS and generative AI tools, Nov 2024).